Daksh — Class X Study Test

Subject: Science | **Test:** Test 2 (Day 14, 23 May 2026) — 2-Week Cumulative
Time Allowed: 3 Hours • Maximum Marks: 80

Syllabus covered: Chs 1–6: Chemical Reactions and Equations • Acids, Bases and Salts • Metals and Non-metals • Carbon and its Compounds • Life Processes • Control and Coordination

General Instructions:
(i) This question paper contains five sections — A, B, C, D and E.
(ii) Section A has Multiple Choice / Very Short Answer Questions of 1 mark each.
(iii) Section B has Short Answer Questions of 2 marks each.
(iv) Section C has Short Answer Questions of 3 marks each.
(v) Section D has Long Answer Questions of 5 marks each.
(vi) Section E has Case-based / Source-based questions of 4 marks each.
(vii) All questions are compulsory. Internal choice has been provided in some questions wherever indicated.
(viii) Use of calculators is not permitted.

SECTION A — Multiple Choice / Very Short Answer Questions (20 × 1 = 20 marks)

Answer all questions.

Q1. 2H■O → 2H■ + O■ on passing electricity is an example of: (a) combination (b) decomposition (c) displacement (d) double displacement

Q2. A solution turns blue litmus red. Its pH could be: (a) 8 (b) 11 (c) 4 (d) 7

Q3. Galvanisation is a process of coating iron with: (a) tin (b) zinc (c) copper (d) chromium

Q4. Functional group present in ethanoic acid is: (a) –OH (b) –CHO (c) –COOH (d) –CO–

Q5. The next homologue of CH■OH is: (a) C■H■OH (b) CH■CHO (c) HCHO (d) CH■COOH

Q6. The site of photosynthesis in plant cells is: (a) mitochondria (b) chloroplast (c) ribosome (d) nucleus

Q7. Human heart has how many chambers? (a) 2 (b) 3 (c) 4 (d) 5

Q8. The basic unit of the nervous system is: (a) neuron (b) nephron (c) tendon (d) axon

Q9. Which of the following is a plant hormone? (a) Insulin (b) Thyroxine (c) Cytokinin (d) Adrenaline

Q10. The pancreas secretes the hormone: (a) thyroxine (b) adrenaline (c) insulin (d) growth hormone

Q11. Which of the following is the most reactive halogen? (a) F (b) Cl (c) Br (d) I

Q12. Bell metal is an alloy of: (a) Cu and Sn (b) Cu and Zn (c) Fe and C (d) Cu and Ni

Q13. The bond between two carbon atoms in C■H■ is: (a) single (b) double (c) triple (d) ionic

Q14. Which acid is found in stomach gastric juice? (a) HNO■ (b) HCl (c) H■SO■ (d) CH■COOH

Q15. The xylem in plants is responsible for: (a) transport of food (b) transport of water and minerals (c) transport of hormones (d) transport of CO■

Q16. The hindbrain part responsible for body balance is: (a) cerebellum (b) cerebrum (c) medulla (d) pons

Q17. Define an isomer.

Q18. Name the gas that turns lime water milky.

Q19. What is haemoglobin? Why is it important?

Q20. Write the chemical formula of bleaching powder.

SECTION B — Short Answer-I (6 × 2 = 12 marks)

Answer in 30–50 words.

Q21. Why is the reaction of aluminium with iron(III) oxide called the thermite reaction? Write the balanced equation.

Q22. Why are carbon and its compounds used as fuels for most applications? Give two reasons.

Q23. What is double circulation? Why is it necessary in human beings?

Q24. List two differences between aerobic and anaerobic respiration.

Q25. Why does the flow of signals in a synapse from axonal end of one neuron to dendritic end of another neuron occur in only one direction?

Q26. Why is sodium hydrogen carbonate added in baking powder along with the acid?

SECTION C — Short Answer-II (7 × 3 = 21 marks)

Answer in 50–80 words.

Q27. What is meant by oxidation and reduction? Define a redox reaction giving an example with the species oxidised and reduced clearly identified.

Q28. What is the chlor-alkali process? Mention the products formed and one use of each.

Q29. Explain the structure of soap molecule. How does soap clean greasy clothes?

Q30. Describe the structure and working of nephron in our body with the help of a labelled diagram.

Q31. Explain the role of the following hormones: (i) thyroxine, (ii) growth hormone, (iii) adrenaline.

Q32. Why do metals usually displace less reactive metals from their salt solutions? Give an example.

Q33. What is the difference between an unsaturated and a saturated hydrocarbon? Why are unsaturated hydrocarbons more reactive?

SECTION D — Long Answer (3 × 5 = 15 marks)

Answer in detail.

Q34. (a) Name the products formed when ethanol reacts with: (i) sodium metal, (ii) hot concentrated H■SO■ at 443 K. Write balanced equations. (3) (b) Why is ethanol used as a fuel additive? Give two advantages and one disadvantage. (2)

Q35. (a) Draw a labelled diagram of the human respiratory system showing trachea, bronchi, bronchioles, alveoli and diaphragm. (3) (b) Describe the exchange of gases between alveoli and blood. (2)

Q36. (a) What is reflex action? Explain the path of a reflex arc with the help of a flow diagram. (3) (b) Why is the response in nervous system faster than in hormonal system? Compare with one example each. (2)

SECTION E — Case-based / Source-based (3 × 4 = 12 marks)

Read each case and answer the sub-parts.

Q37. ***Case Study (Acids, Bases and Salts).*** Plaster of Paris is used by doctors to set fractured bones. It is also used in making toys, decorative articles and false ceilings.
(i) Write the chemical name and formula of plaster of Paris. (1)
(ii) How is it prepared from gypsum? (Equation) (1)
(iii) What happens when water is added to plaster of Paris? (1)
(iv) Why should plaster of Paris be stored in a moisture-proof container? (1)

Q38. ***Case Study (Carbon and its Compounds).*** A homologous series is a family of organic compounds having the same general formula and similar chemical properties.
(i) Define homologous series. (1)
(ii) Write the general formula of an alkene. (1)
(iii) Write the structural formulae of butane and 2-methylpropane (isobutane). (1)
(iv) Why do isomers have similar but not identical properties? (1)

Q39. ***Case Study (Life Processes).*** Photosynthesis is the process by which green plants prepare their own food using sunlight, CO■ and water.
(i) Write the balanced overall equation of photosynthesis. (1)
(ii) Mention the role of chlorophyll. (1)
(iii) Why are stomata necessary for photosynthesis? (1)
(iv) State two factors that affect the rate of photosynthesis. (1)

— End of Question Paper —